TUTORIAL 4

Linear Dependence, Spanning Sets, and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Motivation. A homogeneous linear constraint allows two admissible states to be added, or an admissible state to be scaled, without leaving the admissible set. Span describes which states can be constructed; independence identifies components that are not redundant. Together, these ideas describe a system using a small collection of independent components.

NOTATION AND CONVENTIONS

Notation guide. $\mathbb{F}$ is the scalar field, V a vector space over it, 0_V the zero vector, and $\alpha \in \mathbb{F}$ a scalar. $v \in S$ denotes membership and $S \subseteq V$ inclusion. $\operatorname{span}_{\mathbb{F}} S$ is the span over the indicated field. $U \cup W$ denotes union, $U \cap W$ intersection, $U + W$ a sum of subspaces, and $U \oplus W$ a direct sum. The symbol $\setminus$ denotes set difference and $\varnothing$ the empty set.

DEFINITION 4.1

A precise formulation with hypotheses. For $S \subseteq V$, its span consists of all finite sums $\sum_{j=1}^{k} \alpha_j v_j$ with $v_j \in S$ and $\alpha_j \in \mathbb{F}$, including the empty sum, which equals 0_V. It is the smallest subspace containing S. Independence means that every finite zero linear combination of distinct members of S has only zero coefficients. In particular, the empty set is independent and its span is $\{0_V\}$.
For subspaces $U, W \leq V$ over the same field, $U + W$ is the smallest subspace containing both. The condition $U \cap W = \{0_V\}$ is equivalent to every $x \in U + W$ having a unique expression $x = u + w$ with $u \in U, w \in W$. Indeed, two expressions give $u - u' = w' - w \in U \cap W$. Conversely, if $z \in U \cap W$, then $z = z + 0_V = 0_V + z$, so uniqueness forces $z = 0_V$.

NOTE

A vector is an element of the vector space, so a matrix or polynomial can also be a vector. In an independence test, the unknowns are the *coefficients of the combination*; in a span-membership test, the target vector is given. For a nonhomogeneous condition such as $x + y + z = 1$, first check whether the zero vector belongs to the set.

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MOTIVATION AND GUIDING IDEA

Tutorial objectives. Test linear dependence and independence, find spanning sets, and distinguish the union, intersection, and sum of subspaces. State the scalar field in every test and connect the equations with their geometric interpretation.

4.1 Vector Spaces

RECALL AND TOOLS

1. To define a vector space V over a field $\mathbb{F}$, specify addition of vectors in V and multiplication of a vector in V by a scalar in $\mathbb{F}$, satisfying the vector-space axioms. These operations are specified **in addition** to the addition and multiplication in $\mathbb{F}$. Multiplication of two vectors is not a vector-space axiom; any such operation used later must be defined separately.
2. Every field is a vector space over itself, using its own addition and multiplication.
3. A vector space V is always defined over a field $\mathbb{F}$, which must be stated. For example, as discussed in the lecture, $\mathbb{C}$ can be viewed as a vector space over $\mathbb{C}$ or over $\mathbb{R}$. These are two different vector-space structures.

EXAMPLE 4.1

Let $\mathbb{F}$ be a field.

(a) $\mathbb{F}^n$: the set of n-tuples over $\mathbb{F}$ ($n \in \mathbb{N}$).

(b) $M_{m\times n}(\mathbb{F})$: the set of $m \times n$ matrices over $\mathbb{F}$.

(c) $\mathbb{F}[X]$: the set of all polynomials with coefficients in $\mathbb{F}$.

EXERCISE 4.1

Please determine whether the following sets are subspaces of $\mathbb{R}[X]$ over $\mathbb{R}$, with the usual operations.

1. $V = \{p(X) \in \mathbb{R}[X] : p(5) = 0, p(4) \neq 0\}$.
2. $V = \{p(X) \in \mathbb{R}[X] : \exists c \in \mathbb{R}, p(c) = 0\}$.

WORKED SOLUTION

1. No. The zero polynomial is not in the set because its value at 4 is 0. Closure under addition also fails: $p(X) = (X-5)(X-3)$ and $q(X) = (X-5)^2$ belong to the set, since $p(4) = -1$ and $q(4) = 1$, but
$$(p+q)(X) = (X-5)(2X-8), \qquad (p+q)(4) = 0.$$
Thus $p+q$ does not belong to the set.
2. No. The set is not closed under addition. For example, $(X+1)^2, -2X \in V$, but $(X+$

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EXERCISE 4.1 CONTINUED

$1)^2 - 2X = X^2 + 1 \notin V$.

4.1.1 Subspaces

DEFINITION 4.2

Let V be a vector space over a field $\mathbb{F}$. A subset $U \subseteq V$ is a **vector subspace** of V if it is itself a vector space over $\mathbb{F}$ under the addition and scalar multiplication inherited from V.

PROPOSITION 4.1

Let V be a vector space over a field $\mathbb{F}$ and $U \subseteq V$. Then U is a subspace if and only if all the following conditions hold:

1. $0 \in U$, where 0 is the zero vector of V.
2. U is closed under addition.
3. U is closed under scalar multiplication.

EXAMPLE 4.2

1. $V = \{p(X) \in \mathbb{F}[X] : p(5) = 0\}$ is a subspace of $\mathbb{F}[X]$. **Proof.** The zero polynomial belongs to V. If $p, q \in V$ and $r = p + q$, then $r(5) = p(5) + q(5) = 0$, so $r \in V$. If $p \in V$, $\alpha \in \mathbb{F}$, and $r = \alpha p$, then $r(5) = \alpha p(5) = 0$.
2. $V = \{(x_1, \ldots, x_n) \in \mathbb{R}^n : x_2 = 0\}$, where $n \geq 2$, is a subspace of $\mathbb{R}^n$. **Proof.** Clearly $0 \in V$. If $x = (x_1, \ldots, x_n), y = (y_1, \ldots, y_n) \in V$, then $x_2 = y_2 = 0$, so the second coordinate of $x + y$ is zero and $x + y \in V$. If $x \in V$ and $\alpha \in \mathbb{R}$, then $\alpha x_2 = 0$, hence $\alpha x \in V$.

EXERCISE 4.2

Please prove or disprove.

1. The set $U = \left\{ \begin{pmatrix} a \\ a^2 \end{pmatrix} : a \in \mathbb{R} \right\}$ is a subspace of $\mathbb{R}^2$.
2. The set of symmetric matrices $\mathrm{Sym}_n(\mathbb{F}) := \{A \in M_{n\times n}(\mathbb{F}) : A = A^T\}$ is a subspace of $M_{n\times n}(\mathbb{F})$.

WORKED SOLUTION

1. This is not a subspace, since it is not closed under addition. Indeed,

$$\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1^2 \end{pmatrix} \in U, \qquad \begin{pmatrix} 1 \\ 1 \end{pmatrix} + \begin{pmatrix} 1 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 2 \end{pmatrix} \notin U.$$

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EXERCISE 4.2 CONTINUED

2. The zero matrix is symmetric. Let $A, B \in \mathrm{Sym}_n(\mathbb{F})$ and $\lambda \in \mathbb{F}$. Then

$$(A+B)^T = A^T + B^T = A + B,$$

so $A+B$ is symmetric. Also,

$$(\lambda A)^T = \lambda A^T = \lambda A,$$

so λA is symmetric. All three conditions of the subspace test hold.

EXERCISE 4.3

An engineering sensing system models a normalized signal by a real polynomial of degree at most 3:

$$p(t) = a_0 + a_1 t + a_2 t^2 + a_3 t^3.$$

Time t is measured in normalized time units. This is a mathematical approximation to a signal, with no assumption about the sensor type.

1. Please find all polynomials satisfying

$$p(0) = 0, \quad p(1) = 0.2, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

Is the solution set a subspace of $\mathbb{R}[t]$? Is $t = 0.5$ a local maximum or minimum?

2. Repeat the question with the modified measurements

$$p(0) = 0.1, \quad p(1) = 0.3, \quad p'(0) = 0, \quad p'(0.5) = 0.$$

3. Add the measurement $p'(1) = 0.4$ to part 2. Is the system still consistent? Is its solution set a subspace?

WORKED SOLUTION

Differentiate:

$$p'(t) = a_1 + 2a_2 t + 3a_3 t^2.$$

Part 1. The conditions at $t = 0$ give $a_0 = a_1 = 0$, and the remaining conditions give

$$\begin{cases} a_2 + a_3 = 0.2, \\ a_2 + 0.75 a_3 = 0. \end{cases}$$

Subtract the second equation from the first:

$$0.25 a_3 = 0.2 \quad \Longrightarrow \quad a_3 = 0.8, \qquad a_2 = -0.75 \cdot 0.8 = -0.6.$$

There is therefore a unique solution,

$$p_1(t) = -0.6 t^2 + 0.8 t^3.$$

Verification: $p_1(1) = -0.6 + 0.8 = 0.2$ and $p_1'(0.5) = -1.2 \cdot 0.5 + 2.4 \cdot 0.25 = 0$. The solution set is not a subspace: the zero polynomial does not satisfy $p(1) = 0.2$. The condition $p'(0.5) = 0$

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EXERCISE 4.3 CONTINUED

alone identifies a stationary point, not necessarily a peak. Here

$$p_1''(t) = -1.2 + 4.8t, \qquad p_1''(0.5) = 1.2 > 0,$$

so the point is a local minimum.

Part 2. Now $a_0 = 0.1$ and $a_1 = 0$. The condition $p(1) = 0.3$ gives $a_2 + a_3 = 0.2$, so the two equations for a_2, a_3 are the same as in part 1. Thus

$$p_2(t) = 0.1 - 0.6t^2 + 0.8t^3 = p_1(t) + 0.1.$$

Again the solution is unique, and the solution set is not a subspace because it does not contain the zero polynomial. The derivatives of p_1, p_2 coincide, so $t = 0.5$ is again a local minimum.

Part 3. The unique polynomial from part 2 satisfies

$$p_2'(1) = -1.2 + 2.4 = 1.2 \neq 0.4.$$

The added condition is therefore inconsistent with the others. There is no solution, and the empty solution set is not a subspace because it does not contain the zero vector. In contrast, $\{0\}$ is a subspace.

4.2 Linear Dependence and Independence

DEFINITION 4.3

Let V be a vector space over $\mathbb{F}$ and $S \subseteq V$.

1. The set S is **linearly dependent** if there are pairwise distinct vectors $v_1, \ldots, v_k \in S$ and scalars $\alpha_1, \ldots, \alpha_k \in \mathbb{F}$, not all zero, such that
$$\alpha_1 v_1 + \cdots + \alpha_k v_k = 0.$$
2. The set S is **linearly independent** if, for every finite choice of pairwise distinct vectors $v_1, \ldots, v_k \in S$, this equality implies $\alpha_1 = \cdots = \alpha_k = 0$.

The vectors need not be tuples: they may also be polynomials, matrices, or elements of another vector space.

EXAMPLE 4.3

The set

$$\left\{ \begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} \right\}$$

is linearly dependent, since

$$1 \cdot \begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix} - 1 \cdot \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} + 3 \cdot \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix} = 0.$$

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Rearranging this equation expresses $\begin{pmatrix} 0 \\ -3 \\ 1 \end{pmatrix}$ in terms of the other vectors.

How Do We Test Dependence?

MOTIVATION AND GUIDING IDEA

- Write a **homogeneous** system whose unknowns are the coefficients in the linear combination.
- Form its matrix and row-reduce.
- **Linear independence:** the only solution is the trivial, or zero, solution.
- **Linear dependence:** there is a nonzero solution. Over an infinite field there are then infinitely many solutions.
- A homogeneous system always has a solution: row reduction cannot produce an inconsistent row.

EXERCISE **4.4**

Let $v_1 = (1, 2, 0)^T$, $v_2 = (0, 1, 1)^T$, and $v_3 = (2, 1, -3)^T$ in $\mathbb{R}^3$. Please determine whether they are linearly dependent. Please find all dependence relations and please express one vector in terms of the others.

WORKED SOLUTION

Write $av_1 + bv_2 + cv_3 = 0$. Comparing coordinates gives

$$a + 2c = 0, \qquad 2a + b + c = 0, \qquad b - 3c = 0.$$

The first equation gives $a = -2c$, and the third gives $b = 3c$. The second then becomes $-4c + 3c + c = 0$, imposing no further restriction. All dependence relations are

$$(a, b, c) = t(-2, 3, 1), \qquad t \in \mathbb{R}.$$

In particular, $-2v_1 + 3v_2 + v_3 = 0$, hence $v_3 = 2v_1 - 3v_2$. This is a nontrivial relation. The first two vectors are independent: the first coordinate of $av_1 + bv_2 = 0$ gives $a = 0$, and then $b = 0$.

4.2.1 Useful Criteria for Linear Dependence

1. Every set containing the zero vector is linearly dependent.
2. A singleton consisting of a nonzero vector is linearly independent.
3. **Testing a set for linear dependence.**
   - As in the previous tutorial, set a linear combination equal to zero, obtain a homogeneous system, row-reduce, and determine its solutions.
   - Recall: only the zero solution means independence; a nonzero solution means dependence.

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- Another method from the lecture is to place the vectors as the **rows** of a matrix and row-reduce. If a zero row occurs, the original rows are dependent; if there are no zero rows, they are independent.

4.3 Linear Combinations

DEFINITION 4.4

Let V be a vector space over a field $\mathbb{F}$ and let $v_1, \ldots, v_k \in V$. A vector $v \in V$ is a **linear combination** of $v_1, \ldots, v_k$ if there are $a_1, \ldots, a_k \in \mathbb{F}$ such that $v = a_1 v_1 + \cdots + a_k v_k$.

EXERCISE 4.5

Please determine whether $A = \begin{pmatrix} 2 & 5 \\ -1 & -1 \end{pmatrix}$ lies in the span over $\mathbb{R}$ of $B_1 = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$ and $B_2 = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$.

WORKED SOLUTION

For every $a, b \in \mathbb{R}$,

$$aB_1 + bB_2 = \begin{pmatrix} a & a \\ b & b \end{pmatrix}.$$

Equality with A would require both $a = 2$ and $a = 5$, a contradiction. Thus A is not in the span. Matching the second row using $b = -1$ is insufficient: all four entries must match simultaneously.

EXERCISE 4.6

Let $A = \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix}$ over $\mathbb{R}$. Please characterize all matrices B satisfying $AB = BA$, and please show that they are precisely the linear combinations of A and $I_2 - A$.

WORKED SOLUTION

Write $B = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and compute the two products separately:

$$AB = \begin{pmatrix} 1a + 3c & 1b + 3d \\ 0a + 0c & 0b + 0d \end{pmatrix} = \begin{pmatrix} a + 3c & b + 3d \\ 0 & 0 \end{pmatrix},$$

$$BA = \begin{pmatrix} a1 + b0 & a3 + b0 \\ c1 + d0 & c3 + d0 \end{pmatrix} = \begin{pmatrix} a & 3a \\ c & 3c \end{pmatrix}.$$

Entry $(2,1)$ gives $c = 0$, and entry $(1,2)$ gives $b = 3(a-d)$. All remaining equalities then hold automatically. Consequently,

$$B = \begin{pmatrix} a & 3(a-d) \\ 0 & d \end{pmatrix} = a \begin{pmatrix} 1 & 3 \\ 0 & 0 \end{pmatrix} + d \begin{pmatrix} 0 & -3 \\ 0 & 1 \end{pmatrix} = aA + d(I_2 - A).$$

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EXERCISE 4.6 CONTINUED

The converse holds as well: substituting $c = 0$ and $b = 3(a - d)$ in the two products gives the same expression.

4.3.1 Spanning Sets

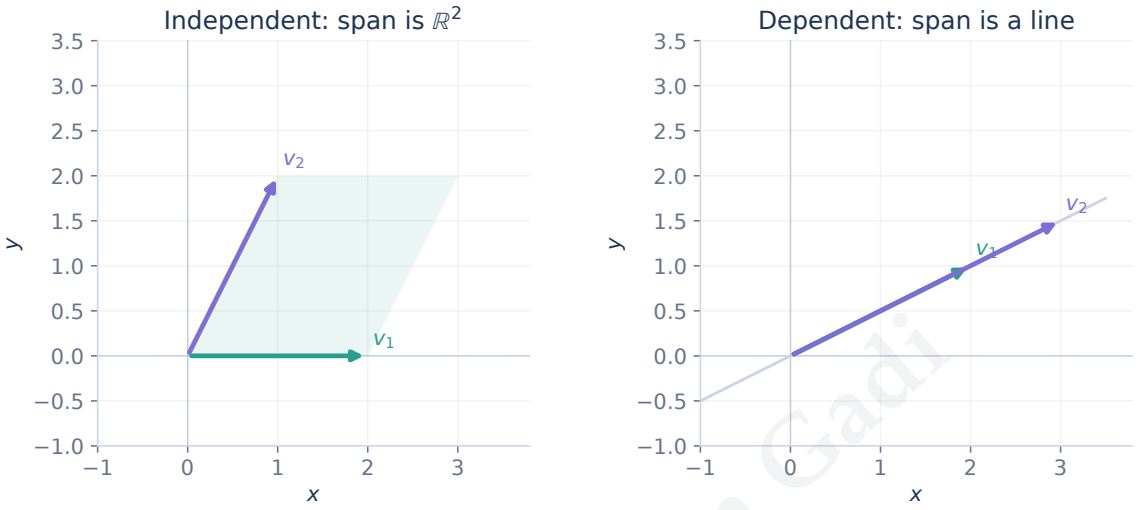


Figure 4.1. Geometric span: a nonzero vector spans a line through the origin; two independent vectors span a plane through the origin.

DEFINITION 4.5

Let V be a vector space over a field $\mathbb{F}$. For a set $S \subseteq V$, not necessarily finite, define $\operatorname{span}_{\mathbb{F}} S$ as the set of all finite linear combinations of elements of S:

$$\operatorname{span}_{\mathbb{F}} S = \left\{ \sum_{j=1}^{k} \alpha_j v_j \;\middle|\; k \geq 0,\ v_j \in S,\ \alpha_j \in \mathbb{F} \right\}.$$

The empty sum ($k = 0$) is allowed and equals the zero vector. In particular, $\operatorname{span}_{\mathbb{F}} \varnothing = \{0\}$.

NOTE

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$, then $\operatorname{span} S$ is a subspace of V.

DEFINITION 4.6

If V is a vector space over $\mathbb{F}$ and $S \subseteq V$ satisfies $V = \operatorname{span} S$, then S **spans** V and is called a **spanning set**. If S is finite, V is **finitely generated**.

NOTE

Pay attention to the scalar field. A set $S \subseteq V$ can span over one field and fail to span over another. When more than one field is possible, write $\operatorname{span}_{\mathbb{F}} S$ to specify the field.

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EXERCISE 4.7

Let V be a vector space over $\mathbb{F}$, let $\varnothing \neq S \subseteq V$ be a finite set of size k, and let $v \in V$. Please prove that

$$\operatorname{span}(S \cup \{v\}) = \operatorname{span} S \quad \Longleftrightarrow \quad v \in \operatorname{span} S.$$

WORKED SOLUTION

$\Longrightarrow$: Assume $\operatorname{span}(S \cup \{v\}) = \operatorname{span} S$. Write $v = 0 \cdot v_1 + \cdots + 0 \cdot v_k + 1 \cdot v$. Thus v is a linear combination of elements of $S \cup \{v\}$, so $v \in \operatorname{span}(S \cup \{v\})$. The assumed equality gives $v \in \operatorname{span} S$.

$\Longleftarrow$: The inclusion $\operatorname{span}(S \cup \{v\}) \supseteq \operatorname{span} S$ always holds: any linear combination of elements of S is also one of elements of $S \cup \{v\}$, with coefficient zero on v. For the opposite inclusion, write $S = \{s_1, \ldots, s_k\}$ and take $x \in \operatorname{span}(S \cup \{v\})$. There are $a_1, \ldots, a_{k+1} \in \mathbb{F}$ with

$$a_1 s_1 + \cdots + a_k s_k + a_{k+1} v = x.$$

Since $v \in \operatorname{span} S$, there are $b_1, \ldots, b_k \in \mathbb{F}$ with $b_1 s_1 + \cdots + b_k s_k = v$. Substituting gives

$$(a_1 + a_{k+1} b_1) s_1 + \cdots + (a_k + a_{k+1} b_k) s_k = x.$$

Hence $x \in \operatorname{span} S$, as required.

EXERCISE 4.8

Please find a spanning set for the solution space of each homogeneous system in $\mathbb{R}^3$:

1. $\begin{cases} x + y = 0, \\ x + z = 0. \end{cases}$
2. $y + z = 0$.

WORKED SOLUTION

1. The second equation gives $x = -z$, and the first gives $y = -x = z$. Set $z = t \in \mathbb{R}$. Then

   $$(x, y, z) = (-t, t, t) = t(-1, 1, 1),$$

   and therefore

   $$S_1 = \{t(-1, 1, 1) : t \in \mathbb{R}\} = \operatorname{span}_{\mathbb{R}}\{(-1, 1, 1)\}.$$

   Every multiple of the given vector satisfies both equations, so this is the entire solution set.

2. From $y + z = 0$ we obtain $y = -z$, while x is free. Set $x = s$, $z = t$, where $s, t \in \mathbb{R}$. Then

   $$(x, y, z) = (s, -t, t) = s(1, 0, 0) + t(0, -1, 1).$$

   Thus

   $$S_2 = \operatorname{span}_{\mathbb{R}}\{(1, 0, 0), (0, -1, 1)\}.$$

   Both vectors satisfy the equation, and every solution has the required form.

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4.4 Intersections and Sums of Subspaces

MOTIVATION AND GUIDING IDEA

Distinguish union from sum. For example, the two coordinate axes in $\mathbb{R}^2$ are subspaces, but their union is not closed under addition: $(1,0)$ and $(0,1)$ belong to the union, whereas their sum $(1,1)$ does not. The sum of the two axes is the whole plane.

RECALL AND TOOLS

If $U, W \subseteq V$ are subspaces of a vector space V, their **intersection** is

$$U \cap W = \{v \in V : v \in U \text{ and } v \in W\}.$$

DEFINITION 4.7

For subspaces $U, W \subseteq V$, define their **sum** by

$$U + W = \{u + w : u \in U, w \in W\}.$$

Then $U + W$ is a subspace of V containing both U and W. If also $U \cap W = \{0\}$, the sum is called a **direct sum** and is denoted $U \oplus W$.

EXERCISE 4.9

Please write

$$\mathrm{Sym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = A^T\}, \qquad \mathrm{aSym}(\mathbb{F}) = \{A \in M_n(\mathbb{F}) : A = -A^T\}$$

for the symmetric and skew-symmetric matrices. Please prove that both sets are subspaces of $M_n(\mathbb{F})$. In addition, please prove that if $\mathrm{char}(\mathbb{F}) \neq 2$, that is, $1 + 1 \neq 0$ in the field, then

$$M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) \oplus \mathrm{aSym}(\mathbb{F}).$$

WORKED SOLUTION

We verify that $\mathrm{Sym}(\mathbb{F})$ is a subspace. For $\mathrm{aSym}(\mathbb{F})$, the zero matrix belongs to the set, and if $A^T = -A$, $B^T = -B$, and $\lambda \in \mathbb{F}$, then

$$(A + B)^T = A^T + B^T = -A - B = -(A + B), \qquad (\lambda A)^T = \lambda A^T = -\lambda A.$$

Thus skew-symmetric matrices are also closed under addition and scalar multiplication and form a subspace.
Now let $A, B \in \mathrm{Sym}(\mathbb{F})$ and $\lambda \in \mathbb{F}$. First, $0 \in \mathrm{Sym}(\mathbb{F})$. For addition, $(A + B)^T = A^T + B^T = A + B$, so $A + B$ is symmetric. For scalar multiplication, $(\lambda A)^T = \lambda A^T = \lambda A$, so λA is symmetric. To prove the direct-sum assertion, first prove that the sum is the whole matrix space and then prove that the intersection is zero. For the equality $M_n(\mathbb{F}) = \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$, check both inclusions:

1. $M_n(\mathbb{F}) \supseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$ holds because both summands are subspaces of $M_n(\mathbb{F})$.

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EXERCISE 4.9 CONTINUED

2. For the reverse inclusion, take $A \in M_n(\mathbb{F})$ and write

$$A = \frac{A + A^T}{2} + \frac{A - A^T}{2}.$$

The first summand is symmetric, since

$$\left(\frac{A + A^T}{2}\right)^T = \frac{A^T + (A^T)^T}{2} = \frac{A + A^T}{2},$$

and the second is skew-symmetric, since

$$\left(\frac{A - A^T}{2}\right)^T = \frac{A^T - (A^T)^T}{2} = \frac{A^T - A}{2} = -\frac{A - A^T}{2}.$$

Hence $M_n(\mathbb{F}) \subseteq \mathrm{Sym}(\mathbb{F}) + \mathrm{aSym}(\mathbb{F})$.

Finally, suppose $A \in \mathrm{Sym}(\mathbb{F}) \cap \mathrm{aSym}(\mathbb{F})$. Symmetry gives $A = A^T$, while skew-symmetry gives $-A = A^T$. Thus $A = -A$, and $a_{ij} = -a_{ij}$ for every $1 \le i, j \le n$. Consequently $a_{ij} + a_{ij} = 2a_{ij} = 0$. Because $\mathrm{char}(\mathbb{F}) \neq 2$, the scalar $2 = 1 + 1$ is nonzero and invertible in the field. Therefore every $a_{ij} = 0$, so $A = 0$, as required.

RECALL AND TOOLS

In characteristic 2, $-A = A$, so the conditions $A^T = A$ and $A^T = -A$ coincide. This is exactly why the symmetric/skew-symmetric decomposition using division by 2 requires characteristic different from 2.

4.5 Supplementary Exercises: Span, Redundancy, and the Scalar Field

EXERCISE 4.10

Three signal components satisfy the balance constraint $x + y + z = 0$. Let

$$U = \{(x, y, z) \in \mathbb{R}^3 : x + y + z = 0\},$$

$$u = (1, -1, 0), \quad v = (0, 1, -1), \quad w = (1, 0, -1).$$

Please show that $U = \mathrm{span}_{\mathbb{R}}\{u, v\}$, please determine whether u, v are independent, and please check what changes when w is added to the spanning set. Does $x + y + z = 1$ define a subspace?

WORKED SOLUTION

Every combination $au + bv = (a, -a + b, -b)$ has coordinate sum zero, hence belongs to U. Conversely, if $(x, y, z) \in U$, then $y = -x - z$, so

$$(x, y, z) = (x, -x - z, z) = x(1, -1, 0) - z(0, 1, -1) = xu - zv.$$

The two inclusions prove $U = \mathrm{span}\{u, v\}$. If $au + bv = 0$, the first coordinate gives $a = 0$, and the third gives $-b = 0$. Thus the pair is independent. Since $w = u + v$, adding w leaves

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EXERCISE 4.10 CONTINUED

the span unchanged and introduces the relation $u + v - w = 0$. The constraint $x + y + z = 1$ does not define a subspace because the zero vector does not satisfy it. It describes a translated plane parallel to U.

EXERCISE 4.11

Consider $S = \{1, i\} \subseteq \mathbb{C}$. Test linear dependence and spanning first over $\mathbb{R}$, then over $\mathbb{C}$.

WORKED SOLUTION

Over $\mathbb{R}$, if $a + bi = 0$ with $a, b \in \mathbb{R}$, comparing real and imaginary parts gives $a = b = 0$. Thus S is independent. Every $z = x + iy$ can be written $x \cdot 1 + y \cdot i$, so $\text{span}_{\mathbb{R}} S = \mathbb{C}$. Over $\mathbb{C}$, complex coefficients are allowed, and

$$i \cdot 1 + (-1) \cdot i = 0.$$

This is a zero linear combination with nonzero coefficients, so S is dependent. It still spans: for every $z \in \mathbb{C}$, $z = z \cdot 1$. In particular, $\{1\}$ alone spans over $\mathbb{C}$.

Additional practice

EXERCISE 4.12 FOUNDATION

Work over $\mathbb{R}$. Let $u = (1, 1, 0)^T$ and $v = (0, 1, 1)^T$.

1. Please describe $U = \text{span}\{u, v\}$ by an equation.
2. Please determine whether $(2, 5, 3)^T$ and $(2, 4, 3)^T$ belong to U, finding coefficients when possible.
3. Please determine whether u, v are independent.

WORKED SOLUTION

For $a, b \in \mathbb{R}$,

$$au + bv = (a, a + b, b)^T.$$

Every vector in the span therefore satisfies $y = x+z$. Conversely, if $(x, y, z)^T$ satisfies $y = x+z$, then

$$(x, y, z)^T = xu + zv.$$

Thus $U = \{(x, y, z)^T : y = x + z\}$. For the first vector, $5 = 2 + 3$, and the vector is $2u + 3v$. For the second, $4 \neq 2 + 3$, so it is not in the span. Finally, if $au + bv = 0$, the first and third coordinates give $a = b = 0$. Thus u, v are independent; in particular, the coefficients found above are unique.

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EXERCISE 4.13 CHALLENGE

Let U, W be subspaces of a vector space V over an arbitrary field $\mathbb{F}$. Please prove that the following are equivalent:

1. $U \cup W$ is a subspace of V.
2. $U \subseteq W$ or $W \subseteq U$.
3. $U + W = U \cup W$.

WORKED SOLUTION

(1) $\Rightarrow$ (2): Suppose neither inclusion holds. Choose $u \in U \setminus W$ and $w \in W \setminus U$. Both belong to the union, so closure under addition gives $u + w \in U \cup W$. If $u + w \in U$, then $w = (u + w) - u \in U$, a contradiction. If $u + w \in W$, then $u = (u + w) - w \in W$, again a contradiction. Hence one inclusion must hold.
(2) $\Rightarrow$ (3): If $U \subseteq W$, then $U \cup W = W$ and $U + W = W$: every such sum belongs to W, and each $w \in W$ equals $0_V + w$. The other case follows by interchanging the subspaces.
(3) $\Rightarrow$ (1): The sum $U + W$ is a subspace, so the equal set $U \cup W$ is also a subspace. The proof divides by no scalar and therefore applies to finite fields and characteristic 2 as well.

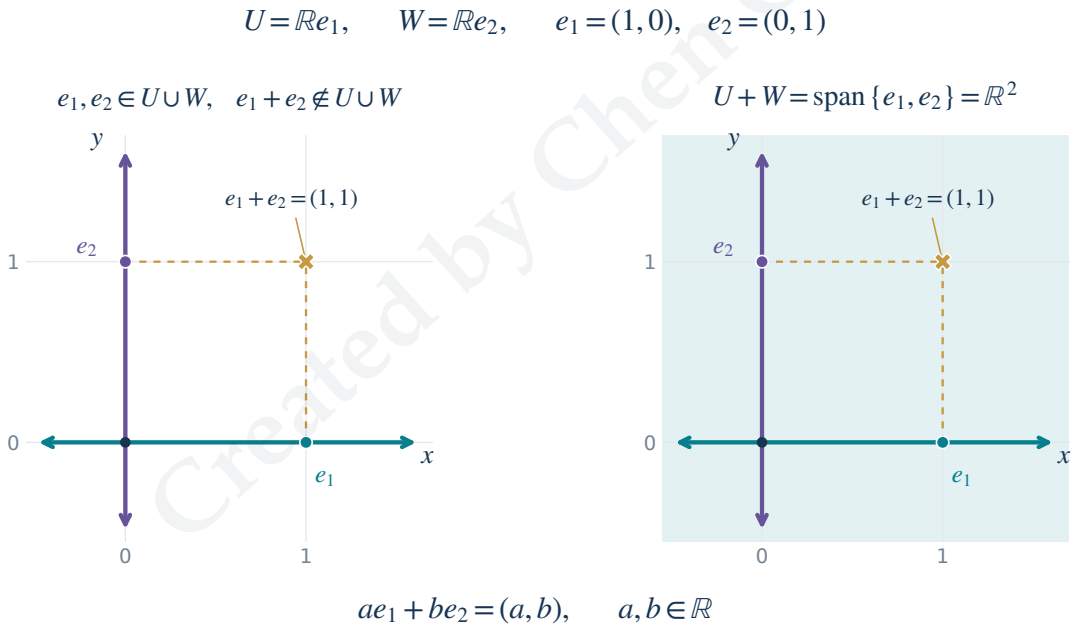


Figure 4.2. An illustration over $\mathbb{R}$: the coordinate axes $U = \mathbb{R}e_1$ and $W = \mathbb{R}e_2$ are subspaces. Their union contains e_1 and e_2 but excludes their sum $(1, 1)$, so it is not closed under addition. In contrast, every $(a, b) \in \mathbb{R}^2$ equals $ae_1 + be_2$, giving $U + W = \mathbb{R}^2$. Shading in the right panel represents the full plane within the viewing window.

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EXERCISE 4.14 CHALLENGE

Let $U_1, \ldots, U_m$ be proper subspaces of a real vector space V, meaning $U_j \neq V$ for every j.

1. Please prove that $U_1 \cup \cdots \cup U_m \neq V$.
2. Please construct three proper subspaces of $\mathbb{Z}_2^2$ whose union is the whole space.

WORKED SOLUTION

We induct on m. The case $m = 1$ is the assumption. By the induction hypothesis, choose

$$a \notin U_1 \cup \cdots \cup U_{m-1}, \qquad b \notin U_m.$$

Consider $a + tb$ for $t \in \mathbb{R}$. For each $j < m$, at most one value of t gives a point in U_j. Two distinct values would imply $b \in U_j$ by subtraction and division by their difference, and then $a \in U_j$, a contradiction. The same line meets U_m in at most one point, since two would imply $b \in U_m$. Only finitely many parameters are excluded, whereas $\mathbb{R}$ is infinite. This proves the claim. Over $\mathbb{Z}_2$, take the spans of $(1, 0)$, $(0, 1)$, and $(1, 1)$. Each is proper, and together they contain zero and all three nonzero vectors of $\mathbb{Z}_2^2$.

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EXERCISE 4.15 CHALLENGE

Let $a_1, \ldots, a_{n+2} \in \mathbb{R}^n$, where $n \geq 1$. For a nonempty index set I, define the convex hull by

$$\operatorname{conv}\{a_i : i \in I\} = \left\{ \sum_{i \in I} \lambda_i a_i : \lambda_i \geq 0,\ \sum_{i \in I} \lambda_i = 1 \right\}.$$

Please prove that the indices can be partitioned into two nonempty sets I, J whose convex hulls intersect.

WORKED SOLUTION

Consider real coefficients $\alpha_1, \ldots, \alpha_{n+2}$ satisfying

$$\sum_{i=1}^{n+2} \alpha_i a_i = 0, \qquad \sum_{i=1}^{n+2} \alpha_i = 0.$$

This is a homogeneous system of $n+1$ equations in $n+2$ unknowns, so it has a nonzero solution. Since the coefficients sum to zero, at least one is positive and at least one is negative. Put

$$I = \{i : \alpha_i \geq 0\}, \qquad J = \{i : \alpha_i < 0\}, \qquad s = \sum_{i \in I} \alpha_i = -\sum_{j \in J} \alpha_j > 0.$$

These sets are nonempty and partition all indices. The vector equation yields

$$\sum_{i \in I} \frac{\alpha_i}{s} a_i = \sum_{j \in J} \frac{-\alpha_j}{s} a_j.$$

On each side the coefficients are nonnegative and sum to 1. The common point therefore lies in both convex hulls.

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